Water Dissolved CO2 sensor Instruction

RD-WDCO2

1. Overview

RD-WDCO2 is a gas detection module based on NDIR infrared absorption principle, which is suitable for detecting carbon dioxide concentration in aqueous solution. RD-WDCO2 adopts patented optical cavity, imported light source and dual-channel detector, which realizes the reference compensation of dual optical paths in space. RD-WDCO2 has the characteristics of good selectivity, no oxygen dependence and long life.

RD-WDCO2 adopts convection diffusion ventilation mode and protective cover, which not only accelerates the convection diffusion speed of gas, but also protects the ventilation film. RD-WDCO2 unique waterproof structure, removable, easy to clean the sensor outer sleeve. RD-WDCO2 can be used to measure the concentration of dissolved CO2 in water, and can also be used to monitor the concentration of CO2 in high temperature and high humidity environment. The output signal is optional, which can meet the requirements of different measurement environments and various high-precision and high-stability measurements.

2. Features

1. High precision infrared carbon dioxide sensor
2. Unique waterproof structure, detachable
3. It is suitable for the concentration of CO2 gas in various liquids
4. Advanced circuit design, high measurement accuracy and stable performance
5. The shell is light and convenient for field installation
6. Rich interfaces, convenient for customers to choose
7. Provide zero calibration, sensitivity calibration and clean air calibration

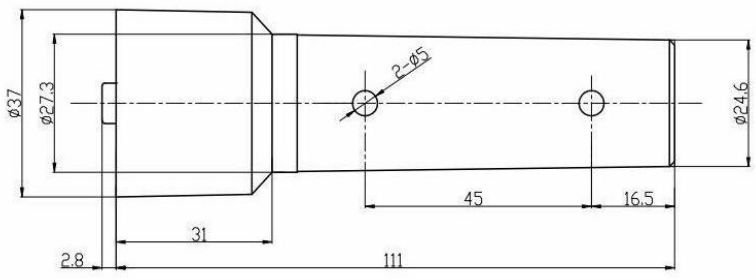
3. Application

1. Aquaculture
2. Water quality monitoring
3. Environmental monitoring of agricultural greenhouses
4. Solution analysis
5. Environmental monitoring of biopharmaceuticals

4. Technical parameters

CO2 CO2	Sensor type: Measurement accuracy: Measuring range: Response time: Calibration:	Infrared principle ± (20PPM +5% reading) 0 ~ 2000PPM < 20s Standard calibration
Output parameter	Signal output:	4 ~ 20mA 0.4 ~ 3VDC UART IIC PWM RS485
485 parameter	Signal type: RS485 Type: Connection: Address: Communication protocol: Data format: Baud rate:	RS485 serial port output Multipoint, half duplex 2-wire system, the maximum communication distance is 1200 meters, and the cable is directly connected 1-254, which is set by software. The broadcast address is 255, and the default address is 32 MODBUS, 1 start bit, 8 data bits, no parity, 1 end bit Default: 9600bit/s
Energy consumption	Power supply:	5VDC ± 5% (RS485, 12V for 4-20mA output)
Wiring mode	Terminal: Cable specification:	None 24 ~ 18AWG (0.5 ~ 1.0 mm), stripping length 2 ~ 3cm
Environment	Working environment:	Temperature-20... + 60 ℃ Humidity 0... 90% RH, no condensation
	Storage environment:	Temperature-20... + 80 ℃ Humidity 0-90% RH, no condensation
Protection level	Shell protection level: Fire rating: Electromagnetic compatibility:	IP68 Refer to IEC 529-598/GB 700-86/GB 4208 V-0 Reference UL94 Refer to GB/T 13926, EN 61000-6-2, EN 61000-6-3
Materials and Colors	Shell: Weight:	ABS (Black) Approximately 100g

5. Overall dimensions



Unit:mm

6. Installation method

Install according to site requirements

Installation Notes:

1. Perform the wiring according to the electrical wiring diagram.
2. After confirming that the wiring is correct, the customer installs it according to the site conditions.
3. After the installation is complete, please conduct a power-on test within the specified power supply range.

7. Electrical wiring

Cable is defined as follows:

UART+IIC wiring definition:

1 Red wire 5V

2 Black wire GND

3 Green wire RXD

4 Yellow wire TXD

5 White wire SDA

6 Brown wire SCL

UART+485 output definition

1 Red wire 12V

2 Black wire GND

3 Green wire RXD

4 Yellow wire TXD

5 White wire 485A

6 Brown wire 485B

UART+4～20mA/DA output definition

1 Red wire 12V

2 Black wire GND& Negative current output & Uart GND

3 Green wire RXD

4 Yellow wire TXD

5 White wire -Positive current output

8. Sensor Communication Protocol

1. Basic communication parameters: baud rate 9600bps, 8-bit data bit, 1-bit stop bit, no parity bit

Sensor default address: 32 (20H)

2. Host Send Protocol Format

The protocol consists of packets in a fixed format. The size of the packet varies according to the content of the message.

Byte	Content
1	Address of communication unit (sensor address)
2	Function code
3	First bit of data area
4	Second bit of data area
5	Third bit of data area
6	Fourth bit of data area
...	Other data
N-1	Low CRC
N	CRC high position

Address of communication unit: This refers to the address of the lower computer unit when the host communicates with the lower computer. The second byte of the packet indicates whether the message is a read command or a write command. 03 indicates that the message is a read command, and 06 indicates that the message is a write command. CRC is used to verify the correctness of transmitted data. Data is transmitted sequentially from low bytes to high bytes. Text is transmitted from left to right. After all data transmission, it will be finished by CRC checking low and high.

Device return protocol format

The protocol consists of packets in a fixed format. The size of the packet varies according to the content of the message.

Command type

(1) Read the sensor concentration value: For example, read the current 32 (20H) sensor data,

The host sends commands to the sensor:

20 03 00 00 00 01 82 BB

20: Current sensor address

03: Read the sensor concentration

00 00: Start register address

00 01: Number of registers

82: High CRC

BB: Low CRC

The basic principle of cyclic redundancy check code (CRC) is to splice R-bit check code after K-bit information code, and the whole code length is N bits, so this code is also called (N, K) code. For a given (N, K) code, it can be proved that there exists a polynomial G (x) with the highest power N-K=R. A check code of K bits of information can be generated according to G (x), and G (x) is called the generation polynomial of this CRC code. The specific generation process of the check code is as follows: assuming that the information to be sent is represented by polynomial C (X), shift C (X) to the left by R bit (which can be expressed as C (X) * 2R), so that R bit will be vacated to the right of C (X), which is the position of the check code. The remainder obtained by dividing C (x) * 2R by the generated polynomial G (x) is the check

code.

The device will return the following data:

20 03 02 01 AB 44 6C (hexadecimal)

20: Current sensor address

03: Read the sensor concentration

02: Data area length

01 AB: Displays sensor gas concentration in PPM

The above data are all hexadecimal numbers, so it is necessary to convert them into decimal numbers before calculating the concentration value

For example:

The feedback data is 01 AB in hex, and the data in decimal number is 427, so the value is 427ppm

44: CRC High

6C: Low CRC

CRC check value reference is the same as above

Note: If you forget the sensor address, please send the FF 03 00 00 00 02 D1 D5, FF is the universal address.

(2) Set the sensor address: For example, change the sensor address No.32 (20H) to No.01

The host sends commands to the sensor:

20 06 00 00 00 01 4E BB (hexadecimal)

20: Current sensor address

06: Function code (set sensor address)

00 00 00 01: Data area (new sensor address 00 01 after modification, which is 01)

4E: CRC High

BB: CRC Low

CRC check value is the same as above

The device will return the following data:

20 06 00 00 00 01 4E BB (hexadecimal)

Consistent with input

After the address is changed, the new read data command only needs to modify the first address to the modified address of the current address and carry out CRC check to obtain a new check bit:

01 03 00 00 00 02 C4 0B (hexadecimal)

The device will return the following data:

01 03 04 00 01 0B E8 AC 8D (hexadecimal)

* This command is the data display window under modbus poll command of serial port debugging assistant. Double-click the address display table to set a new address by modifying value

Note: If you forget the present address and you want to change the sensor address into 01, please send the following:

FF 06 00 00 00 01 5D D4

If success, it will feedback: FF 06 00 00 00 01 5D D4

(3) Set the baud rate

The baud rate can be changed by HEX in RS485 port and the instruction as following

If change into the 4800, please send

2357383131303034354221

If change into the 9600, please send

2357383131303031354521

Note, above instruction is in HEX format and even change successfully, it will not give feedback

And please power off and power on when you changed the baud rate and then it can work.

And above instruction can suitable for any sensor address.

(4) Sensor calibration

A: Make the calibration in the air, place the sensor in an outdoor ventilation environment, stabilize for 10-20 minutes, and then send the following calibration command in hex, the default is 420ppm.

2357353030343230353421

Please pay attention to that, there is no feedback for this instruction and read the data when sending the instruction to check whether it is around 420ppm, if yes, then calibration success.

B: Make the calibration in the standard solution, for example, the standard solution is 1000ppm, stabilize for 10-20 minutes, and then send the following calibration command in hex

2357323030303130303030353421

Please pay attention to that, there is no feedback for this instruction and read the data when sending the instruction to check whether it is around 1000ppm, if yes, then calibration success.

(6) UART protocol

Interface definition: 4-conductor 2.54-pitch socket

Baud rate: 9600bps, 8 data bits, 1 stop bit, no check digit;

The concentration is read and returned as base 16 data;

Zero calibration and SPAN calibration, command and data are ASCII, the number of data bytes per frame is not fixed, 0x20 Beginning, 0x20 end.

The sensor actively uploads the concentration value, and the data is output in ASCII code form in the following format:

For example, if the present value is 723PPM, the format as following

20 20 37 32 33 20 70 70 6D 0D 0A

(7) IIC protocol

The module operates in the slave mode of the IIC and can be connected to an external MCU with internal pull-up resistors. The slave address of the module device is: 0x31

Slave address format:

BIT7	BIT6	BIT5	BIT4	BIT3	BIT2	BIT1	BIT0
0	1	1	0	0	0	1	R/W Bit

R/W Bit: Read = 1/Write = 0

The address of the module for a write operation is: 0x62 and the address for a read operation is: 0x63.

Host send timing:

1）Send start signal

2）Send address write (slave address + R/W Bit(0) = 0x62) and check answer

3）Send read command (ASCII 'R': 0x52)) and check for answer

4）Send stop signal

5）Send start signal

6) Send address read (slave address + R/W Bit(1) = 0x63) and check answer

7）Read 7 bytes from the module and send an answer

	CO2 concentration		Reservations	Reservations	Reservations	Reservations
0x08	High concentration bytes	Low concentration bytes	0x00	0x00	0x00	0x00

9. Considerations

-In environments with EMC problems, shielded protective cables must be used.

-When the module is cold started, the warm-up time shall not be less than 2 minutes.

-The sensor should be calibrated periodically, not more than 3 months is recommended.

-Do not use the sensor for a long time in an environment with high dust density.

-Please pay attention to the wiring of positive and negative poles of power supply and signal output terminal, and do not reverse connection.

-High concentrations of chemical contamination can cause damage to sensors. Therefore, products should be kept and used away from chemical contamination sources.

-Please refer to the relevant national product installation standards for the installation process.